EARTH SCIENCE SPECIAL FEATURE

The Shaking Earth: Causes Of Earthquakes

- UNIT 2 READING RESOURCE
- AUSTRALIAN CURRICULUM ALIGNED
- LESSON 25.2

Earth feels solid under our feet. However, our planet's crust (the hard, rocky outer layer of Earth) is actually broken into huge pieces. These pieces are called tectonic plates (giant puzzle pieces that make up Earth's outer shell). Tectonic plates float very slowly on a softer layer of hot, melted rock deep inside the Earth. They move only a few centimetres each year.

As these plates float, they interact in three main ways. First, some plates pull apart from each other. Second, other plates bump into each other. This bumping can push the land up to form giant mountains. Third, some plates slide sideways past one another. The areas where these plates meet are called boundary zones.

Tectonic Dynamics:
Tectonic plates floating on a softer, molten rock layer beneath Earth's outer crust. Arrows indicate the different ways plates interact: pulling apart, pushing together, or sliding sideways.

The edges of tectonic plates are rough and jagged. As the plates try to move, their rough edges can get stuck together. They stick because of friction, which is a force that resists movement. Even though the edges are stuck, the rest of the plates do not stop moving. They continue to push and pull. This build-up of force creates tension (growing pressure that is stored in the rocks along the boundary).

> *"Even though the edges are stuck, the rest of the plates do not stop moving... This build-up of force creates tension."*
>
> — *TECTONIC MECHANICS*

KEY TERMINOLOGY

Epicentre
The point on the surface directly above the underground start of the earthquake.

Fault Line
The crack in the Earth's crust along which rocks can move.

Focus
The deep underground point where rocks first fracture and slip.

Seismic Waves
Circular energy ripples spreading through the ground.

Seismograph
A special machine that measures the strength of ground movements.

Tectonic
Relating to the structure of the Earth's crust and the large-scale movements of the plates that form it.

Tectonic Plate
A giant puzzle piece of the Earth's hard, rocky outer crust.

Tension
Growing pressure that is stored in rocks when tectonic plates get stuck.

Over time, this pressure becomes too great. The rocks suddenly break or slip. This sudden movement usually happens along a fault (a crack in the Earth's crust where rocks can move). When the rocks suddenly slip, they release a massive amount of stored energy.

Energy Released:
The aftermath of a historical earthquake, demonstrating how massive amounts of stored tension can fracture solid brickwork and split paved roads along active fault lines.

This energy travels outward from the break in all directions. It moves as seismic waves (powerful ripples of energy that travel through the ground). These waves make the ground shake. This shaking is what we feel as an earthquake. The point deep underground where the rocks first broke is called the focus. Directly above this point, on the surface of the Earth, is the epicentre (the point on the surface directly above where the earthquake started). The shaking is always strongest near the epicentre. Smaller shakes called aftershocks (smaller earthquakes that happen after the main shaking) can occur for days or weeks.

Figure 1: Labeled Cross-Section of an Earthquake

[User Image Insertion Area: Labeled diagram showing focus, epicentre, fault line, and seismic waves]

Figure 1 - Earthquake Cross-Section:
A cross-sectional view of an active earthquake zone, highlighting the focal point underground, the surface epicentre, and the propagating circular energy waves.

QUICK FACTS

- ✦ Tectonic plates move at approximately the same rate that human fingernails grow (a few centimetres a year).
- ✦ Friction between plates holds them in place, causing tension to build until the rock structurally fails.
- ✦ Aftershocks can continue for weeks after the main tremor, representing the settling of rock layers.

Scientists measure earthquakes using a seismograph (a special machine that measures the strength of ground movements). By studying earthquakes, we learn how tectonic plates move and how to build safer buildings to protect people.

Measuring Shaking:
A close-up of a seismograph machine translating raw seismic energy waves into measurable line ripples on paper.

Plate Boundary Types (Field Comparison Table)

Boundary Type	Direction of Movement	Geological Features Produced
Convergent Boundary	Plates push into each other	Fold mountains, deep ocean trenches, and strong earthquakes
Divergent Boundary	Plates pull apart from each other	Rift valleys, volcanic activity, and mild earthquakes
Transform Boundary	Plates slide sideways past each other	Active fault lines and shallow, destructive earthquakes